

Physics and Astronomy Department
The University of North Carolina at Greensboro
1400 Spring Garden Street
Greensboro, North Carolina 27412

February 26, 2026

Applying for the Position of Temporary Instructor of Physics

Dr. Pratap,

I write regarding the exciting opportunity to work as a temporary instructor of Physics I and II at UNC-Greensboro, as advertised via announcement. I am currently a doctoral candidate at UNC-Chapel Hill; I have earned my Master's of Science in Physics and have completed all requirements for my doctorate except my dissertation, which I anticipate defending in Spring 2027. As a Royster Doctoral Fellow with non-service funding, I am available to fill the listed position this spring.

My extensive experience as a physics instructor includes a long history of working with students enrolled in introductory physics courses; I have acted as a teacher's assistant at both Wake Forest and UNC-Chapel Hill for these courses. My time teaching was formative, orienting me towards the efficacy of an experiential learning process where students develop an enhanced understanding of curricula via interactive modules. Additionally, I have eight years as a private physics tutor, which has provided me with first-hand insight into the diverse learning styles of students from all walks of life.

I believe that it is crucial that students see the connection between their curricula and how that knowledge can be applied to what is most important in life. As an educator, I benefit from years of philosophical study, which helps give me a deeply grounded, functionalist approach to physics focused on how theory can be applied dynamically to understand and positively shape the world around us. My educational experiences are also buttressed by years of experimental biophysics research, which gives a broad applied knowledge base, empowering me to teach with expertise and nuance. Recently, I mentored a team of undergraduates, guiding them to build a 3D-printed fluorescent microscope capable of advanced particle tracking from inexpensive, commercially available parts. The team went on to win awards for their work at consecutive annual UNC MakerFest competitions; we are now cooperating with low-income high schools in Eastern NC to conduct these builds locally, giving them access to microscopy tools which would normally cost tens of thousands of dollars. These experiences have prepared me to help students understand that studying physics is about more than lab experiments and a curricular requirement, instead offering an empowering understanding of daily life.

Thank you for your time, and I look forward to talking with you about this exciting opportunity in the coming weeks.

Sincerely,

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🌐 philosophiaephysica.org

Royster Fellow | Doctoral Candidate | UNC-Chapel Hill

Education

Doctorate of Philosophy in Physics

Chapel Hill, NC

The University of North Carolina at Chapel Hill

2027

Royster Doctoral Fellow | Exams Passed | All But Dissertation

Master of Science in Physics

Chapel Hill, NC

The University of North Carolina at Chapel Hill

2025

Completed *en route* to PhD

B.S. in Physics, B.A. in Philosophy, Math Minor.

Winston-Salem, NC

Wake Forest University

2022

GPA 3.91 | *Summa Cum Laude* | Stamps Scholar

Academic Projects

Doctoral Thesis.....

Anomalous Diffusion in Poroelastic Media

Advisor: Richard Superfine

Thesis Committee: Dr. Richard Superfine, Dr. C.A. Forrest, Dr. David Hill, Dr. Amy Oldenburg, Dr. Ehssan Nazockdast

Description: Explore the role of static and dynamic strains in the transport of large solutes in various poroelastic media, including transcription factor search patterns in cellular nuclei exposed to dynamic strains.

Masters Thesis.....

Physiological Strains in Poroelastic Media Induce Asymmetric Transport

Advisor: Richard Superfine

Thesis Committee: Dr. Richard Superfine, Dr. Greg M. Forrest, Dr. David Hill, Dr. Amy Oldenburg, Dr. Kerry Bloom

Description: Tracked quantum dots undergoing dynamic diffusion in poroelastic media using LSM with unprecedented temporal resolution. Live imaging during physiological strain reveals that large solute transport is asymmetric during dynamic strains, with advection yielding significantly more forward angular anisotropy during the relaxation stroke than the compression stroke. These results suggest single-particle effects could explain the macroscopic solute concentration enhancements reported by studies of poroelastic media under dynamic strain.

Undergraduate Honors Thesis.....

Generating a 3-D Lattice of Photonic Spheres to Track Fluorescently Labelled Chromatin

Advisor: Dr. Keith Bonin

Committee: Dr. Keith Bonin, Dr. George Holzwarth, Dr. Stephen Baker

Description: Wrote a custom iterative routine to generate, test, and analyze hundreds of thousands of phase settings on a spatial light modulator. Used this routine to create 3D lattice of photonic spheres to be used for tracking labelled chromatin in irradiated cells.

Additional Research.....

Histone Mobility Dynamics Elucidate Chromatin Structural Changes **Chapel Hill, NC**
UNC-Chapel Hill, Dr. Kerry Bloom 2025

Tracking and analyzing the trajectories of HALO-tagged histones during metaphase and interphase to probe structural changes occurring in chromatin.

Enabling Low-Resource Microscopy **Chapel Hill, NC**
UNC-Chapel Hill, Dr. Richard Superfine 2023-Present

Supervised a primarily undergraduate-driven effort to develop cutting edge microscopy techniques for use in low-research environments. Undergraduates won prizes at annual UNC MakerFest competition for two consecutive years.

Employing Advanced LSFM for Enhanced Imaging **Chapel Hill, NC**
UNC-Chapel Hill, Dr. Richard Superfine 2022-Present

Engineered an advanced optical system to conduct new methods of LSFM, such as HILO and HIST. Due to software rot, had to rebuild system from the ground up, involving extensive design and engineering. Also implemented new optical modules and DAQ-board mediated software to enable sub-millisecond automated control of optical hardware.

Synthesizing Chalcogenide Crystals for Quantum Computing **Winston-Salem, NC**
Wake Forest Nanotech Center, Dr. David Carroll 2021

Assisted with the synthesis of chalcogenide crystals being investigated as a potential substrate for quantum computing.

Neurochemical Addiction Mechanisms In the Nucleus Accumbens **Winston-Salem, NC**
Wake Forest School of Medicine, Dr. Evgeny Budygin 2019-2021

Conducted cyclic voltammetry on rats *in vivo* and mouse brain slices *in vitro* to explore the role of neurotransmitters in addictive behaviors.

Cultural Impact of Oenology in the Peloponnese **Winston-Salem, NC**
Wake Forest Department of Anthropology, Dr. Karen Friederic 2019

Conducted a month-long anthropological study in Nemea, Greece on how the economic significance of the viticultural industry impacts local culture. I found that extreme seasonal workloads intersect with traditional gendered domestic labor expectations to inhibit female career growth in this industry.

Cancer and Postoperative Pain **Winston-Salem, NC**
Wake Forest Baptist Medical Center, Dr. Christopher Peters 2018

Prepared solutions, assisted with operations on research subjects, and conducted fluorescence microscopy on brain slices from rats to ascertain the distribution of optogenetic proteins in neurons after treatment with a viral vector.

Scholarships and Grants

2022-2027: Royster Doctoral Fellowship. Five-year fellowship awarded for academic merit.

2018-2022: Stamps Scholarship. Four-year scholarship awarded for academic merit.

2019: Wake Forest Scholars Travel Grant. Funded summer anthropology project abroad.

Publications

Strain alters large solute transport in poroelastic media

Manuscript In Progress, UNC-Chapel Hill

Alexander Marshall, Richard Superfine

Chapel Hill, NC

Anticip. 2026

Dynamic strains drive asymmetric advection in poroelastic media

Manuscript In Progress, UNC-Chapel Hill

Alexander Marshall, Richard Superfine

Chapel Hill, NC

Anticip. 2026

Technical skills

Coding Languages: MATLAB, LabVIEW, Python, L^AT_EX, Git

Scripting: μ Manager, ImageJ

Data Analysis: Particle Tracking, Image Analysis, Stochastic Processes

Other: Transmission Microscopy, Fluorescent Microscopy, Light Sheet Fluorescent Microscopy, Total Internal Reflection Microscopy, Custom Optical Design, Laser Systems, Spatial Light Modulation, Remote Focusing, Machine Vision, Misc. Lab Skills

Lab Safety: I have been lab safety manager for three years. I am responsible for updating the lab safety plan, ensuring compliance during day-to-day operations, keeping everyone up to date with safety trainings, arranging for hazardous waste pickup, coordinating with EHS to maintain a safe workplace, and conducting regular inspections of critical safety equipment.

Talks and Symposia

Physiological Strains in Poroelastic Media Induce Asymmetric Transport

Master's Defense, UNC-Chapel Hill

Chapel Hill, NC

November 17, 2025

Strain Alters Anomalous Diffusion in Poroelastic Media.

Symposium Talk, Elon University

Elon, NC

October 1, 2025

Diffusion in Strained Poroelastic Media

Thesis Proposal, UNC-Chapel Hill

Chapel Hill, NC

September 16, 2025

Generating a 3-D Lattice of Photonic Spheres

Honors Thesis Defense, Wake Forest University

Winston-Salem, NC

May 2022

Generating a 3-D Lattice of Photonic Spheres

Symposium Talk, Wake Forest University

Winston-Salem, NC

May 2022

Conferences

Strain-Driven Transport In Poroelastic Media

Biophysical Society Meeting, Biophysical Society

San Francisco, CA

February 21, 2026

Strain-Driven Transport In Poroelastic Media <i>13th Carolina Biophysics Symposium, UNC-Chapel Hill</i>	Chapel Hill, NC <i>November 6, 2025</i>
Strain Driven Transport In Poroelastic Media <i>Triangle Cytoskeleton Meeting, Duke University</i>	Durham, NC <i>September 15, 2025</i>
Strain-Driven Transport In Poroelastic Media <i>Royster Global Conference, Royster Society of Fellows</i>	Chapel Hill, NC <i>October 21, 2025</i>
Strain-Driven Transport In Poroelastic Media <i>Triangle Soft Matter Workshop, Duke University Dept. of Physics</i>	Durham, NC <i>May 12, 2025</i>
Strain-Driven Transport In Poroelastic Media <i>8th International Soft Matter Conference, Soft Matter Association of the Americas</i>	Raleigh, NC <i>August 2024</i>
Strain Driven Transport In Poroelastic Media <i>Triangle Cytoskeleton Meeting</i>	Raleigh, NC <i>September 23, 2024</i>
Probing Structural and Force Dynamics during Phagocytosis <i>Triangle Cytoskeleton Meeting</i>	Raleigh, NC <i>September 25, 2023</i>
Various topics <i>Annual Colloquium, UNC Applied Physical Sciences Department</i>	Chapel Hill, NC <i>Annually since 2022</i>
The Cultural Impact of Oenology in the Peleponnese <i>Undergraduate Research Day, Wake Forest University URECA Center</i>	Winston-Salem, NC <i>September 2019</i>

Teaching

Research Mentor <i>Royster Fellow, UNC-Chapel Hill</i>	Chapel Hill, NC <i>Fall 2022-Present</i>
Mentored a diverse group of students at the undergraduate and graduate level, helping them to develop as researchers while working on various projects. My persistent efforts at outreach drove undergraduate recruitment and engagements with the lab, leading to a dramatic expansion of the lab's engagement with undergraduates. A summary of select student researchers for whom I served as the primary advisor are included in the mentorship section.	
Teaching Assistant <i>Physics I, Wake Forest University</i>	Winston-Salem, NC <i>Fall 2021</i>
Due to COVID-19 restrictions, the instructor of record was unable to acquire a work visa and could not enter the United States. In her absence, I took on additional responsibilities for the course.	
Teaching Assistant <i>Physics I, UNC-Chapel Hill</i>	Chapel Hill, NC <i>2023-2024</i>
Primarily acted as a lab instructor. Responsible for day-to-day administration of two 110 minute 75 student lab sections which met biweekly. Day-to-day administration of course conducted by TAs. I taught this course three times.	
Teaching Assistant <i>Other Teaching Duties, UNC-Chapel Hill</i>	Chapel Hill, NC <i>2023-2024</i>
Served as a TA for various other courses. Topics covered included Thermodynamics, Quantum Physics II, a first-year seminar centered on the philosophy of science, working in the departmental tutoring center, and more.	

Departmental tutor*Physics I and II, UNC Naval Armory*

Provided physics tutoring for ROTC-enrolled students as part of a departmental initiative.

Chapel Hill, NC*Fall 2023-Spring 2024***Private Tutor***STEM-related Topics*

Worked one-on-one with students of high-school to college age on topics ranging from high school chemistry to Physics Olympiad preparation.

Various Locations*2018-Present*

Mentorship

Ashleigh Smith*Undergraduate Mentee, Anomalous Diffusion*

Ashleigh created a PAAm gel protocol to meet complex experimental requirements for a poroelastic substrate in an isolated chamber. As she has matured as a researcher, this work has segued into a more autonomous project studying the wicking effect of poroelastic media.

Chapel Hill, NC*2024-Present***Mason Miller***Undergraduate Mentee, Anomalous Diffusion*

Mason designed a 3D printed strain control system and ran many strain experiments in poroelastic media. Mason is now working to characterize the rheologic properties of poroelastic media during strains via atomic force microscopy.

Chapel Hill, NC*2025-Present***Bradley Mankus***Undergraduate Mentee, Anomalous Diffusion*

Bradley has improved upon the PAAm gel preparation protocol, has also begun work on developing novel microfluidics channels for testing the effect of advective flows on diffusion in poroelastic media.

Chapel Hill, NC*2025-Present***Shivam Patel***Undergraduate Mentee, Anomalous Diffusion*

Shivam has worked to expand my data analysis codebase, and is currently working to simulate strain-induced advection in poroelastic media.

Chapel Hill, NC*2025-Present***Kautilya Chopra***Undergraduate Mentee, Anomalous Diffusion*

Kautilya created python-based demos of brownian motion from first principles. He then expanded the simulation to model particle trapping and other dynamics typical of diffusion in poroelastic media.

Chapel Hill, NC*Summer 2025***Akshay Elango***Undergraduate Mentee, Low-Resource Microscopy*

Akshay built a 3-D printed transmission microscope based on the OpenFlexure system. His reserch is currenty on hiatus while he works in an industrial internship, but he will return upon its completion.

Chapel Hill, NC*2023-Present***Anya Frazer***Undergraduate Mentee, Low-Resource Microscopy*

Anya expanded upon the OpenFlexure Microscope system, enhancing the capabilities and software of the fluorescence scope, including upgrading it to conduct single particle tracking of nanoscopic particles. She later developed a 3-D printed molecular stamper designed to manufacture Janus particles at dramtically reduced cost in low-resource settings.

Chapel Hill, NC*2024-2025***Eric Chen***Undergraduate Mentee, Low-Resource Microscopy***Chapel Hill, NC***2023-2024*

Eric built a 3-D printed fluorescence microscope based on the OpenFlexure system. He also designed and manufactured his own filtersets for this system from bulk sheets of Schottglass to minimize costs to advance the low-resource intensity goals of the project.

Joshua Adekunle

Undergraduate Mentee, Low-Resource Microscopy

Josh built a 3-D printed low-cost transmission microscope based on OpenFlexure.

Chapel Hill, NC

2023-2024

Outreach

Open Flexure Microscope and Molecular Stamper

UNC-Chapel Hill

Initiatives planned for building microscopes in low-resource highschools in Eastern NC.

Chapel Hill, NC

2023-Present

North Carolina Science Festival

Morehead Planetarium and Science Center

Each year, during this public outreach event the Superfine research group conducts science demos and research talks targeted at general audiences.

Chapel Hill, NC

2023-Present

Hospital Volunteer

Wake Forest Baptist Medical Center

Volunteered in various patient-facing positions for over 300 hours.

Winston-Salem, NC

2015-2017

References

Richard Superfine, PhD

superfine@unc.edu, (919) 432-4381

Taylor-Williams Distinguished Professor, UNC-Chapel Hill

Doctoral Advisor

2022-Present

Kerry Bloom, PhD

kerry_bloom@unc.edu, (919) 962-1182

Thad L. Beyle Distinguished Professor, UNC-Chapel Hill

Reasearch Collaborator

2025-Present

David Hill, PhD

dbhill@email.unc.edu, (919) 962-4718

Associate Professor, UNC-Chapel Hill

Doctoral Committee Member

2025-Present

Keith Bonin, PhD

bonin@wfu.edu, (336) 758-4962

Vice Provost, Wake Forest University

Undergraduate Advisor

2019-2022

Martin Guthold, PhD

guthold@wfu.edu, (336) 758-4977

Physics Dept. Chair, Wake Forest University

Personal Reference

2010-Present

Languages

English: Fluent

Native Speaker

Modern Greek: Intermediate

Conversational speaker.

Classical Latin: Intermediate

Seven years of formal instruction